Ekkati Laxman Reddy

laxmanreddyekkati333@gmail.com | +91 9392428259 | linkedin.com/in/laxmanreddyekkati | github.com/Laxmanredd | Live: datagroq-ai

EDUCATION

Lovely Professional University, Phagwara, Punjab – *B.Tech, Computer Science & Engineering* | *CGPA: 8.1* Aug 2022 – Present

Relevant Coursework: Data Structures & Algorithms, Operating Systems, Database Management Systems, Machine Learning, NLP, Software Testing

Sri Chaitanya Junior College, Hyderabad – *Intermediate* | *96.8%* Apr 2021 – May 2022

INTERNSHIP

Programmer Analyst Trainee – *Cognizant Technology Solutions (GenC Program)* Jan 2026 – Present

- Built and shipped an end-to-end test automation backend — designing a **fault-tolerant Selenium TestNG framework** with structured exception handling, try-catch at every layer, WebDriverWait synchronization, and Assert-based validation to cover edge cases and failure modes beyond the happy path.
- Integrated with external REST APIs using **REST Assured** — writing and validating GET, POST, PUT, PATCH, DELETE requests with JSON/XML schema validation, status code assertions, authentication flows (Basic, OAuth2, Bearer token), and data-driven parameterization via CSV and JSON.
- Worked across a full backend-aligned stack — Java, Maven, GIT/GitHub, Jenkins CI/CD, Docker containers, BDD Cucumber, and Agile sprint workflows — collaborating with senior engineers on system design, service orchestration, and iterative delivery under real deadlines.

SKILLS

Languages:	Python (primary — backend & AI/ML), Java, SQL, Kotlin
Backend & APIs:	REST APIs, REST Assured, Postman, JSON/XML, GraphQL (familiar), Webhooks, API integration & validation
AI & LLM:	Groq API, LangChain, LlamaIndex, Prompt Engineering (Zero-shot, Few-shot, CoT), NLP Pipelines, Scikit-learn
Infra & DevOps:	Docker, GIT/GitHub, Jenkins CI/CD, Maven, BDD Cucumber, Selenium WebDriver, TestNG, POM
Data & Tools:	SQL, NumPy, Pandas, Matplotlib, Tableau, EDA, Hugging Face Spaces, Gradio, Streamlit

PROJECTS

DataGroq – LLM-Powered Document Analysis Backend *[Live]* *Groq API, Python, NLP, REST, Gradio, HuggingFace* | *Sept 2025*

- Designed and deployed a backend LLM pipeline integrating **Groq API** for real-time conversational Q&A over **200+ documents** — engineered with input validation, safe Python code execution guardrails, and structured error handling to ensure robustness beyond the happy path.
- Built end-to-end NLP processing pipelines (tokenization, lemmatization, classification, keyword extraction) with modular, testable components — applying a testing mindset to validate pipeline outputs at each stage and handle edge cases in unstructured data inputs.
- Integrated external APIs and deployed the full backend on Hugging Face Spaces — **reducing manual analysis time by 70%** and improving accuracy by **30%** — with a Gradio frontend layer decoupled from backend logic for clean service separation.

Sentiment Strategy Comparison – ML Pipeline & API Evaluation *Python, TextBlob, Naive Bayes, Prompt Engineering, Gradio* | *Dec 2024*

- Built a rigorous ML evaluation pipeline comparing **6 sentiment strategies** — TextBlob, Naive Bayes, Zero-shot, One-shot, Few-shot, and Chain-of-Thought prompting — on **1,000+ labeled reviews**; applied systematic testing across failure modes achieving **96.8% accuracy** with Naive Bayes.
- Applied prompt engineering techniques (CoT, few-shot) to probe LLM reasoning quality and expose failure modes in generative AI approaches — making deliberate engineering tradeoffs on accuracy vs latency vs interpretability with documented rationale for each strategy.
- Deployed a live Gradio interface decoupled from backend pipeline logic; visualized comparative results via radar charts and bar graphs — delivering an end-to-end system with clear separation of data ingestion, model inference, and presentation layers.

Traffic Analysis Dashboard – Multi-API Data Pipeline *Python, Geoapify REST APIs, Google Sheets API, Tableau* | *Jan–Feb 2025*

- Engineered a backend data pipeline integrating **multiple external REST APIs** (Geoapify, Google Sheets) with automated extraction, transformation, and loading into a dynamic Tableau dashboard — **reducing manual data effort by 80%** with zero manual intervention post-deployment.
- Applied Python-based preprocessing to standardize and validate inputs from heterogeneous API sources — building in data consistency checks and error handling to ensure reliable pipeline output contributing to a **15% increase in operational efficiency**.
- Designed the pipeline with fault-tolerance in mind — handling API response failures, rate limits, and malformed data gracefully; structured the ETL flow with modular components to allow independent testing and easy extension to additional data sources.

CERTIFICATIONS & ACHIEVEMENTS

Introduction to Machine Learning – NPTEL (Oct 2024) | **Data Visualization with Tableau** – Coursera (Nov 2024) | **Python, Java & SQL** – HackerRank (Jan 2025)

Achieved **5-Star rating in SQL** on HackerRank (Feb 2025) | Active Open Source Contributor & Hackathon Participant